

## Safety Guidelines

Life threatening injuries can happen in the laboratory. For that reason, students need to be informed of the correct way to act and things to do in the laboratory. The following is a safety checklist that can be used as a handout to students to acquaint them with the safety do and don't in the laboratory:

1. Wear your lab coats before you enter the lab area
2. Keep the bench top clean, do not stack empty and unused bottles on your working area
3. Disinfect your working area with the proper disinfectant
4. Learn the location of fire extinguishers and how to use them.
5. Know the location of the emergency phone and doors.
6. Keep your laboratory workspace and equipment clean before and at the end of the laboratory session.
7. Wear approved eye protection (i.e., goggles).
8. Notify your instructor immediately in case of an accident.
9. Use disinfectant gel (HiGeen) before and after the experiment.
10. Wear disposable gloves when handling hazardous materials. Remove the gloves before exiting the laboratory.
11. Use the laboratory chemical hood, when there is a possibility of release of any unpleasant odor, toxic chemical vapors, dust, or gases. When using a hood, the sash opening should be kept at a minimum to protect the user and to ensure efficient operation of the hood. Keep your head and body outside of the hood face. Chemicals and equipment should be placed at least six inches within the hood to ensure proper airflow.
12. Wear shoes that adequately cover the whole foot; (no sandals or flip-flops).
13. Don't wear short trousers or skirts.
14. Bags, and other personal items must be stored in designated areas, not on the bench tops.
15. Don't wear contact lenses in laboratory.
16. Remove jewelry (especially dangling jewelry).
17. Synthetic finger nails are not recommended in the laboratory; they are made of extremely flammable polymers which can burn to completion and are not easily extinguished.
18. Use a hot water bath to heat flammable liquids. Never heat directly with a flame.
19. Keep your hands away from your face, eyes, mouth, and body while using

- chemicals.
20. Don't taste or sniff chemicals.
  21. Check the label to verify it is the correct substance before using it.
  22. Don't directly smell any vapor or gas, but fan it towards your nose.
  23. Confine long hair.
  24. Don't eat, drink, and smoke in laboratories.
  25. Consider all chemicals to be hazardous unless you are instructed otherwise.
  26. Dispose of chemicals as instructed by your instructor.
- 
27. Dispose any broken glass only in the marked containers, not in the waste-baskets and dispose items contaminated with body fluids in the specified yellow bag.
  28. If chemicals come in contact with your skin or eyes wash with copious amounts of water or eye wash and then consult your instructor.
  29. Never point a test tube that you heat towards your face or your neighbor because it may erupt.
  30. Do not perform any unauthorized experiment.
  31. Always Add concentrated acid to water slowly. Never add water to a concentrated acid, because the heat of solution will cause the water to boil and the acid will spatter.
  32. Never mix or use chemicals not called for in the laboratory exercise.
  33. Avoid rubbing your eyes unless you know that your hands are clean.
  34. Don't pour chemical waste into the sink drains or wastebaskets.
  35. Place chemical waste in appropriately labeled waste containers.
  36. Properly dispose of sharp objects (e.g., syringe needles) immediately in designated containers.
  37. Many common reagents, like ether, alcohols, and acetone are highly flammable. Do not use them anywhere near open flame.

**Reference:**

School Chemistry Laboratory Safety Guide. Department of health and human services  
Centers for Disease Control and Prevention National Institute for Occupational Safety  
and Health.